INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

SBT-DF204: Computer Forensics Case Studies

Case Study 5: DJI Mavic Air Investigation

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Course code	SBT-DF204 Computer Forensics Case Studies
Assessment	Case Study 5: DJI Mavic Air Investigation
Case title	DJI Mavic Air Investigation
Evidence examined	Evidence basisAndroid logical archive, supplied EWF image, analysis outputs and screenshotrecord
Submission date	2/11/2026

CONFIDENTIALITY AND HANDLING

This report documents a training case-study examination. It contains technical observations derived from the supplied evidence and the recorded practical work. Exact GPS coordinates, device serial identifiers and other unnecessary identifying values are intentionally excluded from this publication copy.

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1. Executive Summary

This report records a digital-forensic examination undertaken for SBT-DF204 Case Study 5, involving DJI MavicAir-related data. The examination focused on a supplied Android logical archive containing DJI GO 4 applicationartefacts, flight records and cached media, together with a supplied Expert Witness Format (EWF) imageassociated with the drone's microSD storage.

The Android archive was tested and extracted in Kali Linux. DJI flight-record TXT data was decoded using a DJITXT parsing utility, and two DAT flight logs (FLY057 and FLY058) were converted to CSV using DatCon. Asupplied EWF image was integrity-verified, mounted for read-only examination, and used to recover five MP4video files. The recovered videos and Android cached media were examined through metadata inspection andselected-frame visual review.

The TXT flight record contained 3,055 records spanning approximately 5 minutes and 5 seconds. The DAT filenamed FLY057 contained 5,261 records, with GPS time values spanning approximately 18:50:34 to 18:55:39UTC. FLY058 contained 3,785 records, with available GPS time values spanning approximately 19:00:23 to19:02:08 UTC. FLY058 showed the controller motor-state field as zero in 3,780 records and front motor-speedvalues of zero in 3,775 records; GPS quality indicators were poor or unavailable in some records, so theseobservations should not be treated as definitive proof of the aircraft's physical state.

Five MP4 files were recovered from the EWF image. Their combined duration was approximately 10 minutes 51seconds. Selected frames from DJI_0001.MP4 and DJI_0002.MP4 depict indoor technical or office-likeenvironments; selected frames from DJI_0003.MP4, DJI_0004.MP4 and DJI_0005.MP4 depict open, dry terrainand distant hills. The Android cached video was approximately 4 minutes 1 second long, 1280 × 720 pixels,encoded in H.264. Its SHA-256 hash did not match any of the five EWF-extracted MP4 files, meaning the filesare not byte-identical; this does not exclude related or re-encoded content.

The Android logical extraction also contained six small cached JPG thumbnails. The available examination foundno EXIF capture date or GPS information in those thumbnails. Filename-derived epoch values and filesystemmodification times were not accepted as proof of original capture time.

Key qualification: the EWF image was examined, but a separate raw microSD .001 image was notsuccessfully obtained. The screenshot record shows that the attempted download returned an HTMLDropbox error page rather than a raw forensic image. The report therefore makes no claim that a raw .001image was acquired or examined.

Overall, the examination recovered and analysed multiple DJI-related artefacts and established useful time andactivity observations. It did not establish the exact geographic location, identity of people appearing in sampledframes, or the purpose of the recorded activity. The conclusions are limited to the available supplied evidenceand the selected-frame review documented in the practical record.

2. Case Background and Objectives

The case study concerns the examination of digital artefacts associated with a DJI Mavic Air drone and its DJIGO 4 mobile application environment. The supplied Android logical archive and EWF image were treated asseparate evidence sources. The purpose of the work was to identify relevant flight records and media, preserveintegrity through hashing and verification, extract supported artefacts, and document findings in a reproduciblemanner.

2.1 Objectives

- Establish a structured forensic workspace and keep original evidence separate from working copies.
- Verify the integrity of the supplied Android archive and EWF image using cryptographic hashes and availableverification utilities.
- Identify DJI application data, flight-record TXT/DAT files, cached JPGs and MP4 video artefacts.
- Decode the TXT flight record and convert DAT flight logs into a form suitable for analysis.
- Recover videos from the supplied EWF image and inspect metadata and selected frames.
- Correlate timestamps and observations across sources without overstating what the data proves.
- Prepare a sanitised publication set that excludes precise coordinates and device serial identifiers.

3. Scope and Questions Examined

The examination addressed the following practical questions:

- What DJI-related artefacts were present in the Android logical dataset?
- Could the TXT and DAT flight records be decoded or converted for analysis?
- What activity and timing information was present in the available flight records?
- Could video files be recovered from the supplied EWF image, and what was visible in sampled frames?
- Did the Android cached video match any EWF-recovered video at the byte level?
- What integrity, privacy and acquisition limitations affected the conclusions?

The work did not include attribution of the drone operator, identification of people shown in the media,geolocation of scenes, or a full frame-by-frame review of every video. No incident hypothesis or alleged offencewas supplied in the materials reviewed; accordingly, the report does not infer intent or wrongdoing.

4. Evidence Inventory

IDEvidence itemDescription / examination status

E-01Android_Logical.zipCourse-supplied Android logical archive. ZIP integrity test completed; archiveextracted to the working area. MD5 and SHA-256 recorded.

E-02Android logical extractionExtracted Android_Logical directory, including DJI GO 4 application folders, TXT/DATflight records, cached JPGs and MP4 media.

E-03DJI TXT flight recordDJIFlightRecord_2018-06-19_[14-50-34].txt; decoded to a readable text output usinga DJI TXT parser.

E-04FLY057.DATDJI DAT flight record converted to CSV using DatCon; 5,261 records reported inanalysis.

E-05FLY058.DATDJI DAT flight record converted to CSV using DatCon; 3,785 records reported inanalysis.

E-06DF048.E01Supplied EWF image associated with the drone microSD; integrity verificationcompleted and image mounted read-only for examination.

E-07DJI_0001.MP4–DJI_0005.MP4Five MP4 files extracted from the EWF image; metadata and selected framesexamined.

E-08Android cached MP42018_06_19_14_50_39.mp4; metadata inspected and selected frames reviewed.

E-09Six cached JPG thumbnailsSmall DJI GO 4 cache images; visual inspection and available metadata reviewcompleted.

E-10Raw microSD .001 imageNot acquired. The attempted download returned HTML error content rather than aforensic image.

5. Tools and Examination Environment

The practical work was performed in Kali Linux within a project workspace under the user's home directory. Theworkspace separated original evidence, working copies, tools, outputs, logs, reports and screenshots. The toolsand utilities evidenced in the work include:

- Kali Linux shell utilities: mkdir, cp, find, file, unzip, md5sum, sha256sum, grep and related command-line tools.
- DJI TXT parsing utility (djiparsetxt), cloned and compiled for decoding the DJI GO 4 TXT flight record.
- DatCon 4.3.0 for conversion of DJI DAT flight logs to CSV.
- libewf utilities, including ewfverify and ewfmount, for EWF integrity verification and read-only mounting.
- Sleuth Kit utility icat for extracting files from the mounted forensic image.
- FFmpeg / ffprobe for video metadata inspection and frame extraction.
- Python scripts and command-line analysis for summarising CSV fields, timestamps, flight modes, motor-statevalues and media hashes.

The screenshot record shows compilation warnings from the TXT parser build. The report treats the parser asused for the subsequent decoding task, but the warnings are retained as a tool-quality note; they are notrepresented as proof that every parser output is error-free.

6. Methodology

6.1 Workspace preparation

A case workspace was created with distinct directories for original evidence, working data, screenshots, evidence, reports and tools. The archive was copied into the original-evidence area and extracted into a workingarea. This separation reduces the risk of altering the supplied source while allowing analysis to be performed onworking copies.

6.2 Identification and extraction of Android artefacts

The extracted Android directory was searched for DJI-related application paths and file extensions. Theexamination identified DJI GO 4 flight-record TXT files, DAT files under the DJI flight-record folder, cached JPGthumbnails and MP4 files under the DJI recording folder. These artefacts were selected for decoding, conversion, metadata inspection and visual review.

6.3 Flight-record processing

The TXT flight record was processed with the djiparsetxt utility. The DAT records were converted with DatCon toCSV so that record counts, timestamps, flight modes, GPS fields, controller motor state and selected telemetryvalues could be inspected. Analysis of location fields was conducted in the working environment; precisecoordinates and selected location-bearing columns were excluded from the public GitHub dataset.

6.4 EWF examination

The supplied DF048.E01 EWF image was verified using the available EWF verification utility. The image wasmounted read-only and examined as a FAT32 volume. Video files were extracted using forensic file extractionmethods and then examined with FFmpeg/ffprobe and selected-frame extraction. The extracted videos wereretained as examination outputs.

6.5 Media examination

Video examination used metadata inspection and selected frames/contact sheets. The resulting visual notesdescribe only what is visible in the sampled frames. Sampling does not establish that every frame or the fullduration of each video was examined. Cached JPG thumbnails were reviewed visually and checked for availableEXIF metadata.

6.6 Publication sanitisation

Before publication, the two CSV files were sanitised by removing 15 location- or privacy-sensitive fields, includinglatitude/longitude and selected derived coordinate fields. A serial value in the FLY058 Attribute/Value section wasreplaced with [REDACTED]. Verification logs were generated to record the sanitisation and final privacy checks.The public repository was configured to include only the sanitised output directory and exclude original evidence, working data and private local folders.

7. Acquisition, Integrity and Preservation

7.1 Android logical archive

The Android_Logical.zip archive was identified as a ZIP archive, tested during extraction, and extracted into theworking directory. The recorded hashes for the archive were:

- MD5: 4f7623911818e8dc1ddceee4e26875be
- SHA-256: 9a352b282d832556eb7cb73010ea281f9946d889438ce07d9bbee3b64229285

The extracted working copy was compared with the supplied archive using the available hash and file verificationprocedures. The archive integrity test reported no extraction errors.

7.2 EWF image

The supplied EWF image DF048.E01 was verified. The recorded container hashes were:

- MD5: d1651cfacd5b1827e337ad0a826002ae
- SHA-256: 435b97c0cba3007d40b4ac10686049d1527fa0dc1235d8492a2df3e66d8bdac9

The EWF verification utility reported successful verification. The embedded media hashes recorded duringexamination were MD5 7285a252d6c4582e120e06f3c13861ff and SHA-18bb139ec8196e238d17e729ce9cfe585adfc616e. These embedded-media values are distinct from the hashes ofthe EWF container file.

7.3 Extracted video integrity

The five extracted MP4 files were hashed after extraction. The Android cached video was also hashed andcompared with the extracted EWF videos. The cached video hash differed from the hashes of all fiveEWF-extracted videos. This establishes that the files are not byte-for-byte identical; it does not establish that theirvisual content or underlying recording events are unrelated.

7.4 Preservation and custody qualification

The practical record documents hash verification and separation of original and working directories. However, thematerials reviewed do not provide a complete formal chain-of-custody form showing every transfer, custodian, date/time, seal identifier and authorisation. This report therefore documents the integrity checks performedwithout claiming a complete institutional chain-of-custody record.

8. Examination and Analysis

8.1 Android logical dataset

The Android logical dataset contained the DJI GO 4 application directory and related subfolders. Searchesidentified two DJI flight-record TXT files, two DAT flight records (FLY057 and FLY058), cached JPG thumbnails, an Android cached MP4 recording and other DJI application media/tutorial files. The principal flight-record andmedia artefacts were prioritised for examination.

8.2 TXT flight record

The TXT record was decoded into a readable text output. The decoded record contained 3,055 records andcovered approximately 305.374 seconds, from 18:50:34.869 to 18:55:40.243 as represented in the decodedrecord. The recorded application version was DJI GO 4 version 4.2.16 and the aircraft model was identified asMavic Air.

The decoded summary reported the following flight-mode counts: GPS_Atti 2,867; AutoTakeoff 92; AutoLanding81; and EngineStart 15. The recorded total distance was 444.66 (unit as represented in the source summary), total time 305.3 seconds, maximum height 28.5, maximum horizontal speed 2.91 and maximum vertical speed2.1. The TXT summary reported photoNum 0. These values are reproduced from the available decodedsummary and should be interpreted in the context of the parser's field definitions and source data.

The TXT filename includes a 14:50:34 time component, while the decoded record time is approximately18:50:34. This difference is noted but not independently resolved as a time-zone conversion. The report does notassume a specific offset without corroborating device configuration or acquisition context.

8.3 DAT flight record FLY057

DatCon conversion produced a CSV containing 5,261 records for FLY057. The available GPS time values spanapproximately 18:50:34 to 18:55:39 UTC. Flight-mode values were predominantly GPS_Atti (4,920 records), withAutoTakeoff (170), AutoLanding (137) and ASST_TAKEOFF (33) also recorded. The controller motor-state fieldwas recorded as 1 in 5,259 records and 0 in one record. The available relative-height values rangedapproximately from -12.70 to 22.53 in the converted dataset.

The original analysis included GPS coordinates. For privacy, this report does not reproduce the precisecoordinate values. The public CSV was sanitised to remove direct and selected derived location fields.

8.4 DAT flight record FLY058

DatCon conversion produced a CSV containing 3,785 records for FLY058. Available GPS time values spanapproximately 19:00:23 to 19:02:08 UTC. The controller flight-mode field showed GPS_Atti in 3,748 records andAtti in 32 records. The controller motor-state field was 0 in 3,780 records, while the front motor-speed value waszero in 3,775 records.

Some FLY058 records showed GPS satellite count (numGPS) of zero and dilution-of-precision values of 99.99.These indicators show that GPS positioning quality was poor or unavailable in those records. Although themotor-state and motor-speed values are consistent with an inactive or non-powered

state for most of therecorded samples, they do not independently prove the aircraft's exact physical position or condition.

8.5 Android cached video

The Android cached video was identified as 2018_06_19_14_50_39.mp4. Metadata analysis reported a durationof approximately 240.972 seconds (about 4 minutes 1 second), frame dimensions of 1280 × 720 and H.264 videoencoding at approximately 29.24 frames per second. Selected frames showed a green open landscape, a whiteevent-style tent, people, vehicles, dirt tracks and distant hills.

The available metadata and filename time values are not treated as conclusive capture-time evidence. Inparticular, filesystem modification/access times can reflect copying, extraction or later system activity rather thanoriginal recording.

8.6 Cached JPG thumbnails

Six cached JPG thumbnails were identified under the DJI GO 4 CACHE_IMAGE folder. The images were smallpreview files (approximately 160 × 90 pixels, with one thumbnail reported at 128 pixels in one dimension). Thevisual review identified five indoor/workshop-style views showing a person in a dark or black shirt near a whitework surface and equipment, and one outdoor image showing dry terrain, buildings and distant snow-cappedhills. The person's identity was not established.

No EXIF capture date, GPS coordinate, camera make or camera model was recovered from the six thumbnails.Numeric filename suffixes can be converted into epoch-like dates, but those values alone do not prove theoriginal capture time. Filesystem modification times were not accepted as original capture timestamps.

8.7 EWF image and extracted videos

The supplied DF048.E01 EWF image was mounted read-only after verification. The examined filesystem wasidentified as FAT32. Five MP4 files, DJI_0001.MP4 through DJI_0005.MP4, were extracted from the image andchecked. Their durations were:

Video fileDuration (seconds) approximate duration

DJI_0001.MP460.7941 min 00.8 sec

DJI_0002.MP452.71952.7 sec

DJI_0003.MP4291.0244 min 51.0 sec

DJI_0004.MP4132.6992 min 12.7 sec

DJI_0005.MP4114.0811 min 54.1 sec

Total651.31710 min 51.3 sec

Selected-frame observations were recorded as follows:

- DJI_0001.MP4: sampled frames depict an indoor environment resembling a classroom, computer laboratory orworkshop, with desks, monitors, equipment and a green plant visible in one frame. Image blur/motion limitsdetailed interpretation.

- DJI_0002.MP4: sampled frames depict an office or technical workspace with desks, computers, monitors andat least two people visible in the sampled footage. Their identities, roles and activities are not established.
- DJI_0003.MP4: sampled frames depict an aerial view of open, dry terrain, scrub/grassland, a wash or trackand distant hills.
- DJI_0004.MP4: sampled frames depict similar open dry terrain, a track and distant hills.
- DJI_0005.MP4: sampled frames depict yellow-brown grassland, a dirt road and distant hills; some smallobjects may be present but are not confidently identified.

The video notes are based on selected frames rather than a complete frame-by-frame examination of every clip.The scenes' exact geographic location and the identities of any people shown were not determined.

9. Findings

F-01 - Android DJI artefacts identified

The extracted Android logical dataset contained DJI GO 4 application data, two TXT flight records, two DAT flightrecords, cached JPG thumbnails and an Android cached MP4 video.

F-02 - TXT record decoded

A DJI TXT flight record was decoded into a readable output containing 3,055 records and an approximately305.374-second record span.

F-03 - FLY057 telemetry converted

FLY057 was converted to CSV with 5,261 records and GPS time values spanning approximately18:50:34–18:55:39 UTC. GPS_Atti was the predominant flight mode.

F-04 - FLY058 telemetry converted

FLY058 was converted to CSV with 3,785 records and GPS time values spanning approximately19:00:23–19:02:08 UTC. Motor-state and motor-speed fields were zero in most recorded samples; GPS qualitywas poor in some records.

F-05 - EWF image verified and examined

The supplied DF048.E01 image passed the recorded integrity verification and was mounted read-only. Five MP4files were extracted from it.

F-06 - Video media recovered

The five EWF-extracted MP4 files had a combined duration of approximately 651.317 seconds. Selected framesdepict both indoor technical/workspace scenes and outdoor open terrain.

F-07 - Android cached media identified

The Android cached MP4 was approximately 240.972 seconds long, 1280 × 720 and H.264. Six cached JPGthumbnails were also identified.

F-08 - No byte-identical video match

The SHA-256 hash of the Android cached MP4 differed from each of the five EWF-extracted MP4 files. This rulesout byte-identical copies, not possible visual or content relationships.

F-09 - Public data sanitised

The public CSV files had 15 location/privacy-sensitive fields removed, and a serial value was redacted.Verification logs were created for the sanitisation checks.

F-10 - Raw .001 image not acquired

The attempted raw microSD .001 download returned an HTML Dropbox error page. The raw image wastherefore not available for forensic examination.

10. Cross-Source Correlation and Timeline

The TXT record's decoded time span begins at approximately 18:50:34.869 and ends at approximately18:55:40.243. FLY057 GPS time values span approximately 18:50:34 to 18:55:39 UTC. The close alignmentsupports a temporal relationship between the TXT record and FLY057. This correlation is based on the availabledecoded/converted timestamps; it does not independently prove that the two files are complete duplicates ordescribe every aspect of the same flight.

FLY058 contains a later interval, approximately 19:00:23 to 19:02:08 UTC. Its motor-state and motor-speed fieldsare zero in most records, while some GPS quality fields are poor. The interval is chronologically later thanFLY057, but the available records alone do not establish why the later log exists or the exact physicalcircumstances represented.

The Android cached MP4 filename includes a 14:50:39 time component, while the TXT and DAT records containvalues around 18:50 UTC. A time-zone difference is one possible explanation, but it has not been establishedfrom the evidence reviewed. Metadata timestamps and filenames are therefore treated as supporting contextonly, not definitive time anchors.

Time / periodSourceObservation

18:50:34.869–18:55:40.243Decoded TXT3,055 records; approximately 305.374 seconds.

18:50:34–18:55:39 UTCFLY057 DAT5,261 CSV records; predominant GPS_Atti mode.

19:00:23–19:02:08 UTCFLY058 DAT3,785 CSV records; motor-state/motor-speed fields zero inmost records.

2018-06-19 14:50:39 infilename

Android cached MP4Filename time is not independently established as UTC orlocal capture time.

Selected-frame reviewEWF MP4s and Androidmedia

Indoor/workspace and outdoor terrain imagery observed; noexact location or identity established.

11. Limitations, Constraints and Evidential Cautions

The following limitations affect the strength or scope of the conclusions. They are recorded to make the reporttransparent and reproducible.

- Raw image unavailable: a separate raw microSD .001 image was not obtained. The attempted downloadreturned HTML error content, so only the supplied EWF image was examined.
- Supplied evidence provenance: the Android logical archive and EWF image were supplied as coursematerials. The report records the hashes and verification performed, but does not independently establish theoriginal collection circumstances before delivery.
- Chain-of-custody documentation: a complete formal chain-of-custody record was not present in thematerials reviewed.
- Selected-frame video review: only selected frames/contact sheets were examined. The work does notrepresent a complete frame-by-frame or continuous playback analysis of every video.
- Timestamp interpretation: filenames, filesystem times and metadata fields can reflect timezone settings,copying or later system activity. The discrepancy between the TXT filename time and decoded/DAT time was notindependently resolved.
- GPS quality: some FLY058 GPS fields indicate unavailable or poor-quality positioning. Those records cannotsupport precise location conclusions.
- Location privacy: exact coordinates were intentionally excluded from the report and public repository. Thereport therefore does not provide a route map or precise geographic reconstruction.
- Identity and attribution: people visible in sampled images or video were not identified. No conclusion is madeabout their roles, ownership of the device or operation of the drone.
- Media correlation: the Android cached MP4 is not byte-identical to any of the five EWF-extracted videos.Visual similarities alone are insufficient to establish that they are the same recording or were captured at thesame location.
- Tool validation: the TXT parser compilation produced warnings in the recorded terminal output. The availablerecord does not include an independent validation of every decoded field against a second parser.

These constraints do not invalidate the completed processing steps. They define the boundaries within which theobservations and conclusions should be relied upon.

12. Conclusion

The examination successfully identified and processed significant DJI-related artefacts from the supplied Androidlogical dataset and supplied EWF image. The Android archive was tested and extracted; a DJI TXT flight recordwas decoded; two DAT flight records were converted to CSV; and five MP4 files were recovered from the EWFimage after integrity verification and read-only mounting. The Android cached MP4 and six cached JPGthumbnails were also identified and examined.

The flight records provide a coherent time sequence for the TXT record and FLY057, with a later intervalrepresented by FLY058. The FLY058 motor-state and motor-speed values are zero in most

samples, althoughGPS quality limitations prevent confident physical-location interpretation. The media review documents indoortechnical/workspace scenes and outdoor open-terrain scenes in selected frames, but does not establishidentities, exact locations or intent.

The report's most important qualification is that the raw microSD .001 image was not acquired; only the suppliedEWF image was verified and examined. The findings should therefore be understood as conclusions from theavailable supplied evidence, not from an independently acquired raw microSD image or a complete review ofevery video frame.

13. Recommendations

- Retain the original supplied archive and EWF image in a controlled, read-only evidence location and preservethe recorded hashes with the case documentation.
- If the raw microSD .001 image is required by the case instructions, obtain it through a working authorisedsource and verify its hash before examination; do not treat an HTML download error as an image.
- Complete a formal chain-of-custody record for any future transfer or re-examination.
- If the case requires detailed video interpretation, perform a full playback and/or systematic frame sampling withdocumented sampling intervals and preserve the extracted frame hashes.
- If precise flight-path reconstruction is required and authorised, analyse the original location-bearing telemetryin a controlled environment; publish only appropriately redacted outputs.
- Validate key TXT-derived fields against a second compatible parser or documented DJI field specificationwhere available.
- Preserve the sanitised GitHub publication set separately from original evidence and do not upload unredactedcoordinates, serial identifiers, original media or private working files.

14. Appendices

Appendix A - Key integrity values

EvidenceHash typeValue

Android_Logical.zipMD54f7623911818e8dc1ddceee4e26875be

Android_Logical.zipSHA-2569a352b282d832556eb7cb73010ea281f9946d889438ce07d9bbee3b64229285

DF048.E01 EWF containerMD5d1651cfacd5b1827e337ad0a826002ae

DF048.E01 EWF containerSHA-256435b97c0cba3007d40b4ac10686049d1527fa0dc1235d8492a2df3e66d8bdac9

Embedded media in EWFMD57285a252d6c4582e120e06f3c13861ff

Embedded media in EWFSHA-18bb139ec8196e238d17e729ce9cfe585adfc616e

Android cached MP4SHA-2562ae2a4f186803ec592b0af6284634e592c5079395e99f7df3edd583f8e72ab00

Appendix B - Sanitised publication

The GitHub publication repository contains the sanitised CSV files, analysis summaries, evidence inventory,visual examination summary, final investigation report and privacy-verification logs. The public dataset excludesthe original evidence, working directory and other private folders. The CSV sanitisation removed 15location/privacy-sensitive fields and redacted a serial value. The repository URL is:https://github.com/IfyTechy/SBT-DF204-case-study-5-dji-forensics

Appendix C - Use of AI assistance

AI assistance was used to explain forensic concepts, provide command-line guidance, troubleshoot selectedissues, help interpret selected command outputs and organise the report structure. Practical operationsdescribed in this report are based on the user's recorded terminal outputs and supplied screenshots. The reportshould be checked against the original evidence and course requirements before formal submission.

Appendix D - Selected Practical Evidence Screenshots

The following figures are selected from the 112-page screenshot evidence PDF supplied for this report. Theoriginal page numbers are retained in each caption to allow the reader to locate the full context in the sourceevidence pack.

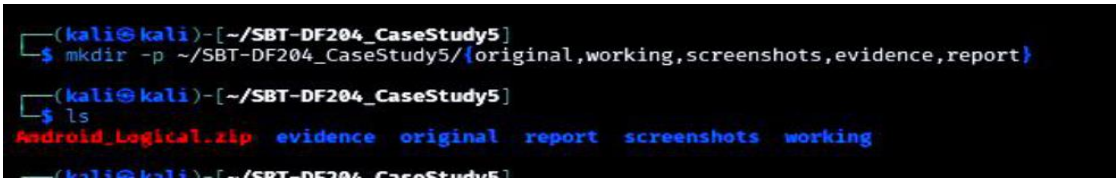


Figure D1.1. Source screenshot page 2: Project workspace created in Kali Linux, with separate original, working, evidence, reportand screenshot directories.

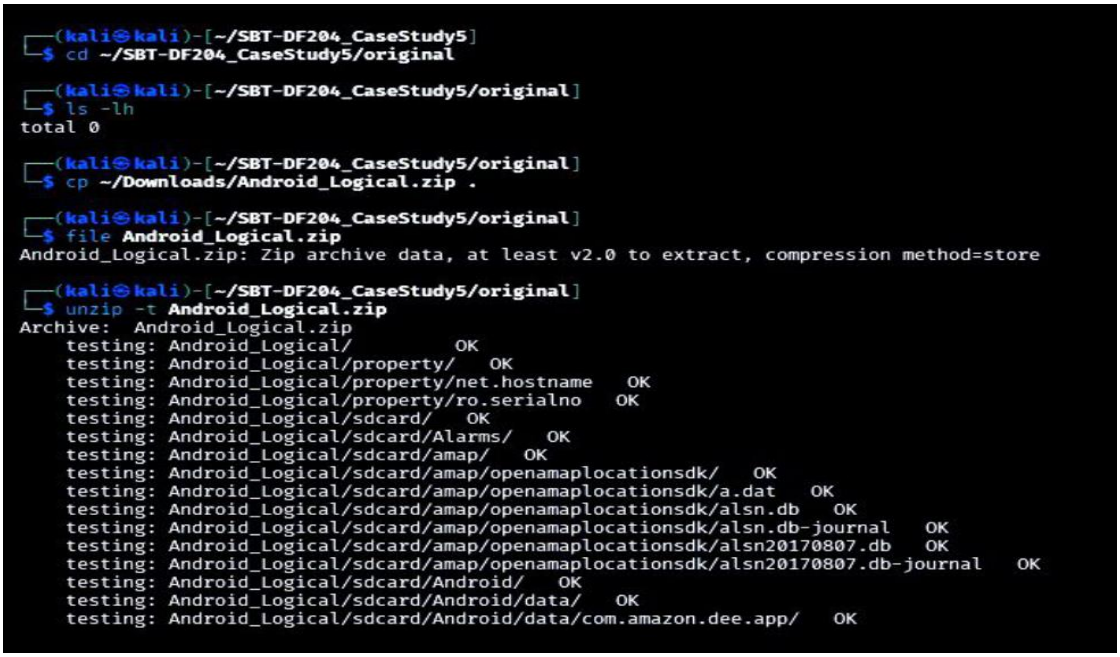


Figure D1.2. Source screenshot page 5: Android logical archive identified as a ZIP file and tested during extraction; the displayedentries report successful testing.

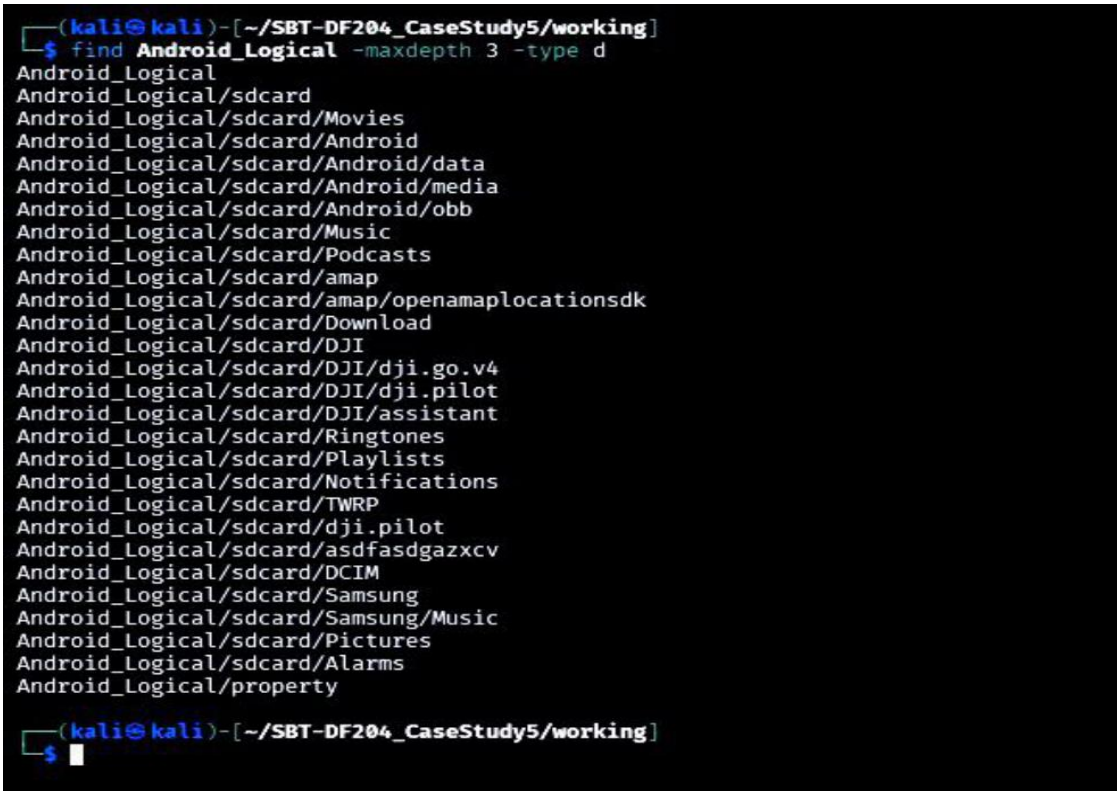


Figure D2.1. Source screenshot page 7: Directory listing of the extracted Android logical dataset, including DJI application dataand media folders.



Figure D2.2. Source screenshot page 9: Identification of DJI GO 4 flight-record TXT and DAT files, cached JPG images and DJIapplication artefacts.

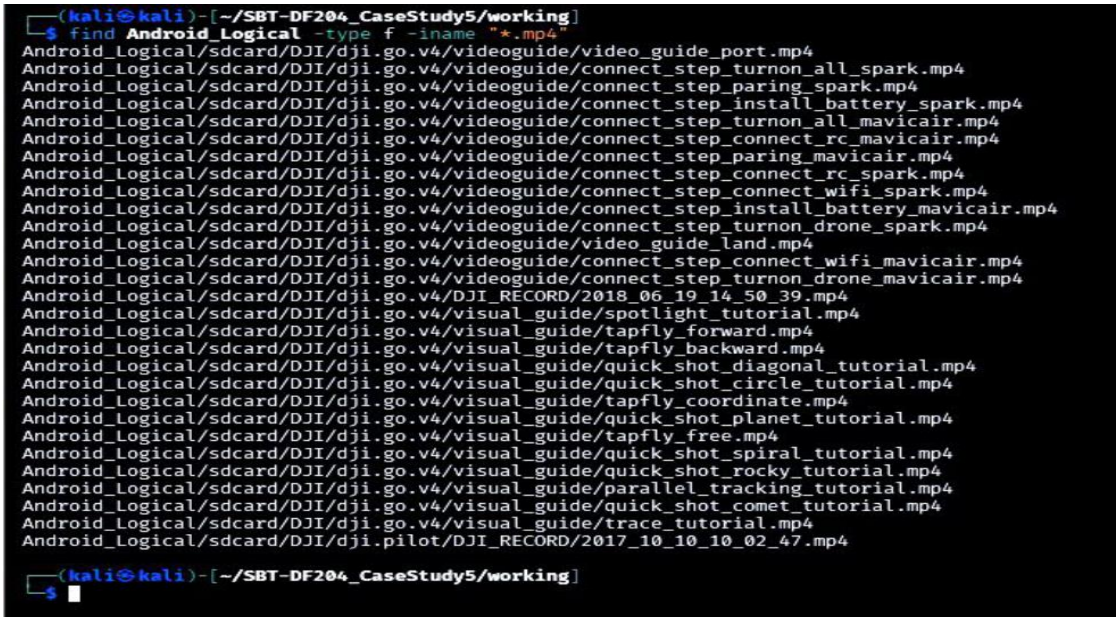


Figure D3.1. Source screenshot page 10: Identification of DJI video files and DJI recording folders within the Android logicaldataset.

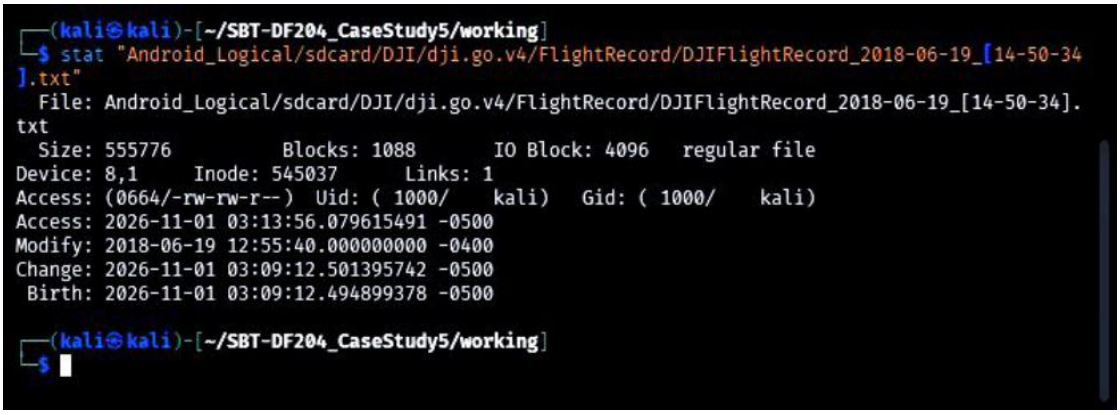


Figure D3.2. Source screenshot page 16: Filesystem metadata displayed for the DJI flight-record TXT file; filesystem timestampsare not treated as original capture timestamps.

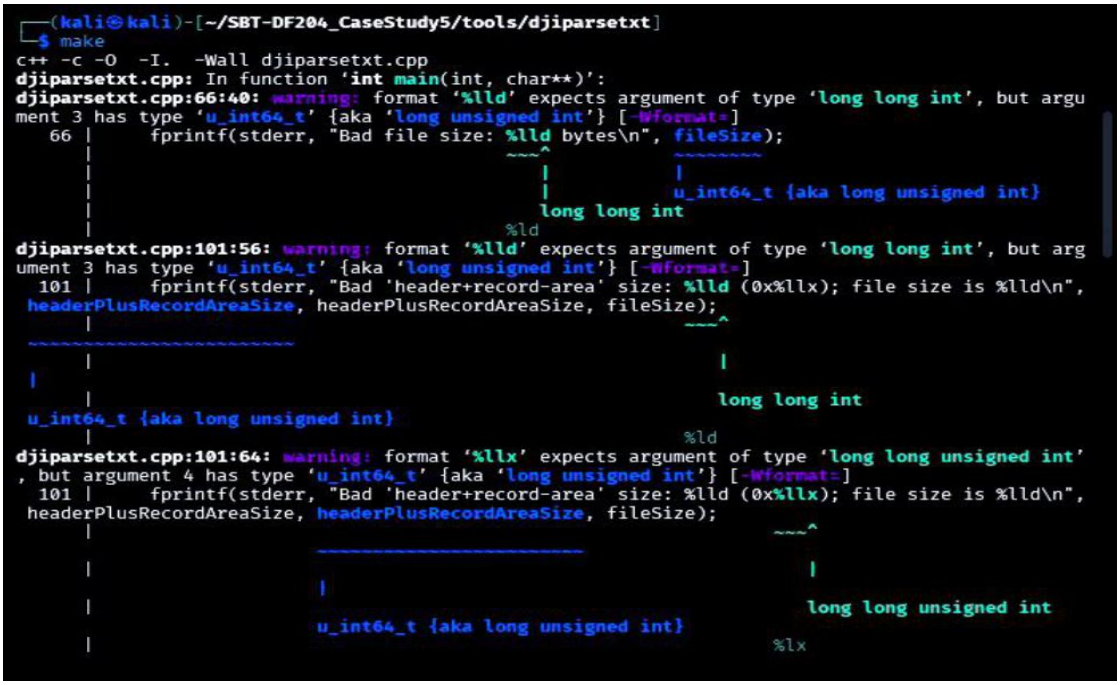


Figure D4.1. **Source screenshot page 18: Compilation output for the DJI TXT parser. Compiler warnings are visible and should bedistinguished from successful decoding.**

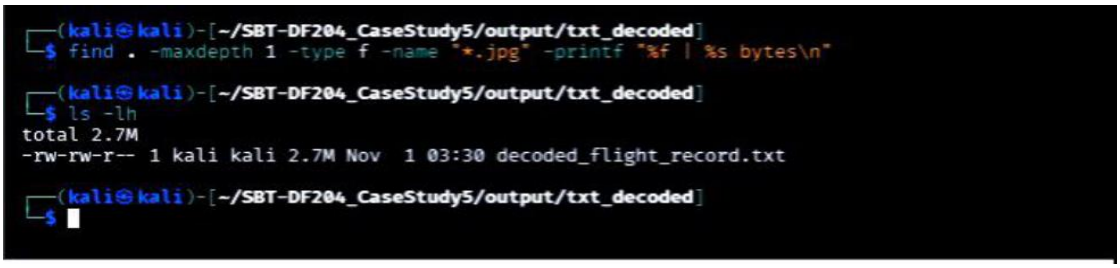


Figure D4.2. **Source screenshot page 29: Decoded TXT flight-record output file present in the project output directory.**



Figure D5.1. **Source screenshot page 35: Hash comparison and copy-verification activity for the two DAT flight-record files.**

```
[sudo] password for kali:
┌──(kali㉿kali)-[~/SBT-DF204_CaseStudy5/tools/DatCon]
└─$ mkdir -p ~/SBT-DF204_CaseStudy5/output/dat_analysis

┌──(kali㉿kali)-[~/SBT-DF204_CaseStudy5/tools/DatCon]
└─$ cd ~/SBT-DF204_CaseStudy5/output/dat_analysis

ls -lah
total 15M
drwxrwxr-x 2 kali kali 4.0K Nov  1 06:05 .
drwxrwxr-x 4 kali kali 4.0K Nov  1 05:54 ..
-rw-rw-r-- 1 kali kali  15M Nov  1 06:05 18-06-19-02-47-38_FLY057.csv

┌──(kali㉿kali)-[~/SBT-DF204_CaseStudy5/output/dat_analysis]
└─$ file *
18-06-19-02-47-38_FLY057.csv: ASCII text, with very long lines (5789)

┌──(kali㉿kali)-[~/SBT-DF204_CaseStudy5/output/dat_analysis]
└─$
```

Figure D5.2. Source screenshot page 41: DatCon output directory and converted FLY057 CSV file.

kali@kali: ~/SBT-DF2.../output/dat_analysis kali@kali: ~/SBT-DF2.../output/dat_analysis

```
        print("First populated GPS record: CSV row", index + 2)
        print({field: row.get(field) for field in fields})
        break

print("\nLAST POPULATED GPS RECORDS:")
count = 0
for index, row in reversed(list(enumerate(rows))):
    if row.get("GPS:Lat", "").strip() and row.get("GPS:Long", "").strip():
        print("CSV row", index + 2,
              {field: row.get(field) for field in fields})
        count += 1
        if count == 5:
            break
PY
First populated GPS record: CSV row 6
{'GPS:dateTimeStamp': '', 'GPS:Lat': '39.9612038', 'GPS:Long': '-106.2162718', 'Controller:ctrl_
mode': '', 'Controller:motor_state:D': '', 'Motor:Speed:RFront': ''}

LAST POPULATED GPS RECORDS:
CSV row 3786 {'GPS:dateTimeStamp': '', 'GPS:Lat': '39.961152', 'GPS:Long': '-106.2162638', 'Cont
roller:ctrl_mode': 'Atti', 'Controller:motor_state:D': '0', 'Motor:Speed:RFront': '0'}
CSV row 3785 {'GPS:dateTimeStamp': '', 'GPS:Lat': '39.961152', 'GPS:Long': '-106.2162638', 'Cont
roller:ctrl_mode': 'Atti', 'Controller:motor_state:D': '0', 'Motor:Speed:RFront': '0'}
CSV row 3784 {'GPS:dateTimeStamp': '', 'GPS:Lat': '39.961152', 'GPS:Long': '-106.2162638', 'Cont
roller:ctrl_mode': 'Atti', 'Controller:motor_state:D': '0', 'Motor:Speed:RFront': '0'}
CSV row 3783 {'GPS:dateTimeStamp': '', 'GPS:Lat': '39.961152', 'GPS:Long': '-106.2162638', 'Cont
roller:ctrl_mode': 'Atti', 'Controller:motor_state:D': '0', 'Motor:Speed:RFront': '0'}
CSV row 3782 {'GPS:dateTimeStamp': '', 'GPS:Lat': '39.961152', 'GPS:Long': '-106.2162638', 'Cont
roller:ctrl_mode': 'Atti', 'Controller:motor_state:D': '0', 'Motor:Speed:RFront': '0'}

┌──(kali㉿kali)-[~/SBT-DF204_CaseStudy5/output/dat_analysis]
└─$
```

Figure D6.1. Source screenshot page 50: Inspection of GPS-populated rows in the converted FLY058 dataset; precisecoordinates are excluded from this report.

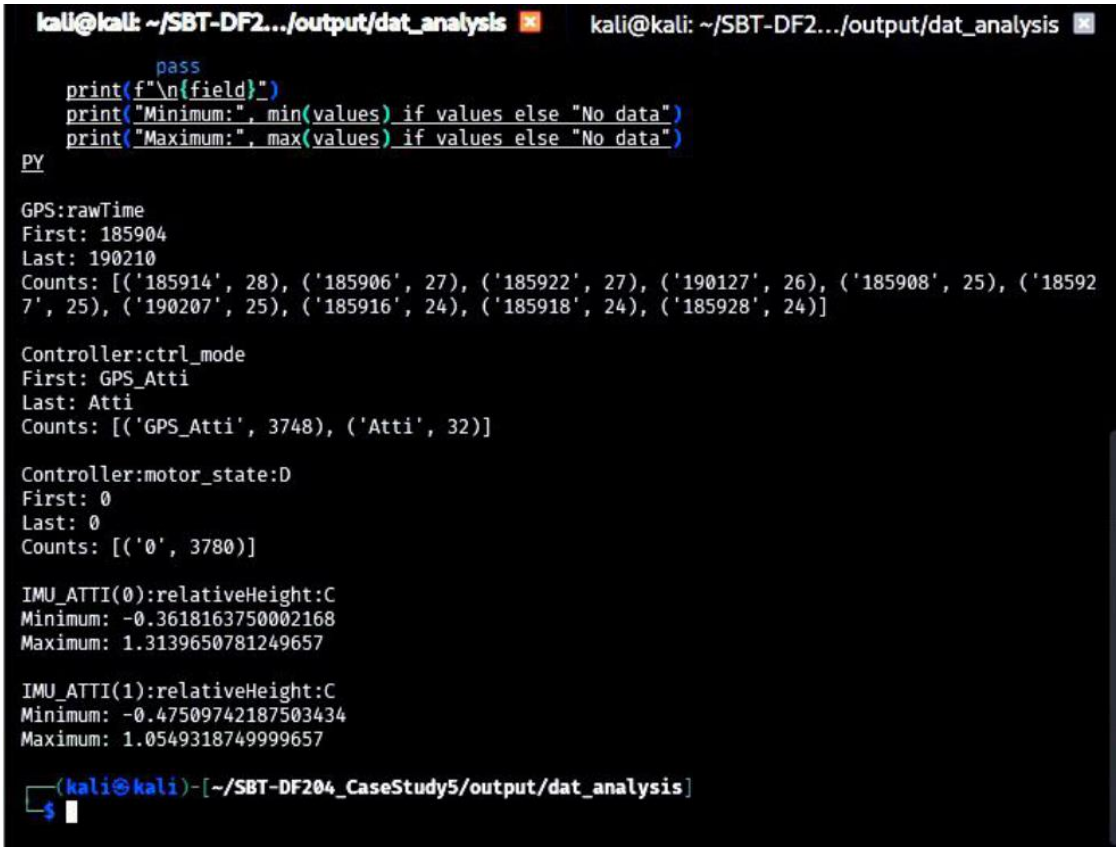


Figure D6.2. Source screenshot page 55: Summary output for FLY058 showing flight-mode and motor-state counts, plusrelative-height values.

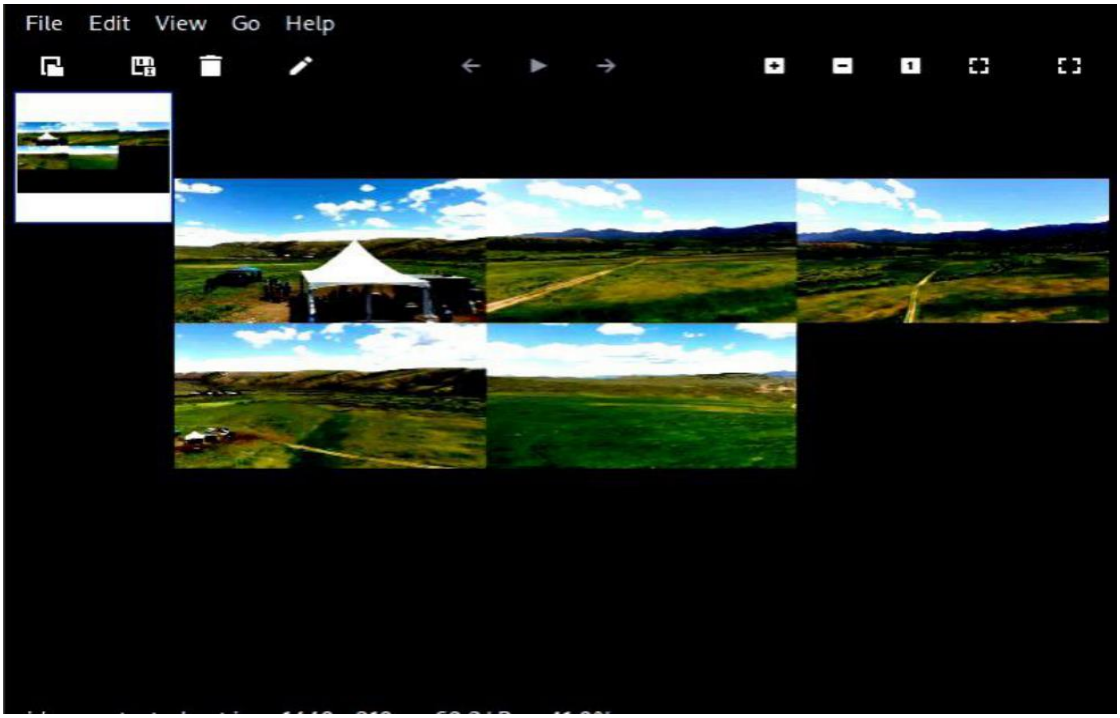


Figure D7.1. Source screenshot page 61: Contact-sheet view of sampled frames from the Android cached video.

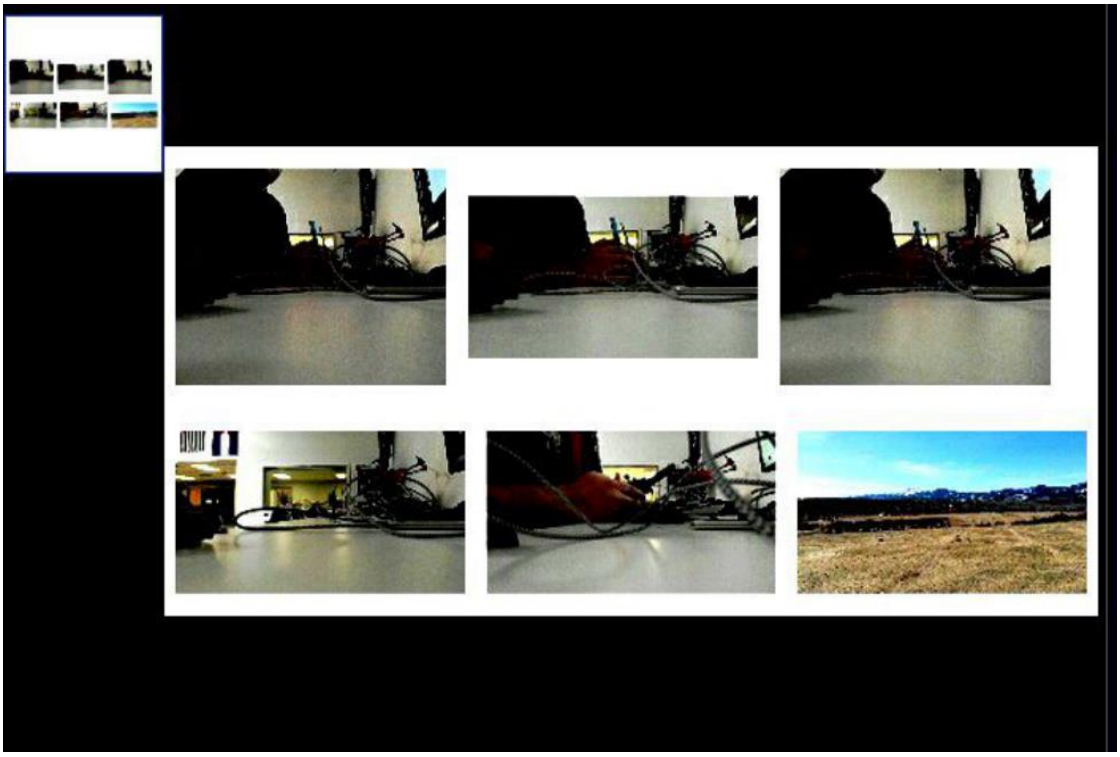


Figure D7.2. Source screenshot page 66: Contact sheet of six cached DJI JPG thumbnails, including indoor/workshop andoutdoor imagery.

```
ly
Output #0, image2, to 'output/ewf_video_frames/DJI_0001/frame_%02d.jpg':
  Metadata:
    major_brand     : isom
    minor_version   : 512
    compatible_brands: isomiso2avc1mp41
    encoder         : Lavf63.1.102
  Stream #0:0(und): Video: mjpeg, yuvj420p(pc, progressive), 640×360, q=2-31, 200 kb/s, 0.10
fps, 0.10 tbn (default)
    Metadata:
      encoder         : Lavc63.1.102 mjpeg
      handler_name    : VideoHandler
    Side data:
      CPB properties: bitrate max/min/avg: 0/0/200000 buffer size: 0 vbv_delay: N/A
frame=    3 fps=0.0 q=1.6 size=N/A time=00:00:30.00 bitrate=N/A speed=0.435x elapsed=0:01:08.
[out#0/image2 @ 0×5649b7a6cdc0] video:144KiB audio:0KiB subtitle:0KiB other streams:0KiB glob
al headers:0KiB muxing overhead: unknown
frame=    6 fps=0.0 q=1.6 Lsize=N/A time=00:01:00.00 bitrate=N/A speed=0.481x elapsed=0:02:04
.68

┌──(kali㉿kali)-[~/SBT-DF204_CaseStudy5]
└─$ ls -lh output/ewf_video_frames/DJI_0001/
total 156K
-rw-rw-r-- 1 kali kali 16K Nov  2 12:06 frame_01.jpg
-rw-rw-r-- 1 kali kali 38K Nov  2 12:07 frame_02.jpg
-rw-rw-r-- 1 kali kali 14K Nov  2 12:07 frame_03.jpg
-rw-rw-r-- 1 kali kali 32K Nov  2 12:07 frame_04.jpg
-rw-rw-r-- 1 kali kali 22K Nov  2 12:08 frame_05.jpg
-rw-rw-r-- 1 kali kali 26K Nov  2 12:08 frame_06.jpg

┌──(kali㉿kali)-[~/SBT-DF204_CaseStudy5]
└─$
```

Figure D8.1. Source screenshot page 90: Extraction and metadata inspection of sampled frames from DJI_0001.MP4.

```
┌──(kali㉿kali)-[~/SBT-DF204_CaseStudy5]
└─$ xdg-open output/ewf_video_frames/DJI_0001/frame_01.jpg

┌──(kali㉿kali)-[~/SBT-DF204_CaseStudy5]
└─$ cat <<'EOF' >> logs/DF048_E01_visual_examination.txt

VIDEO: DJI_0001.MP4
FRAME: frame_01.jpg
SOURCE: Extracted from EWF microSD image DF048.E01

OBSERVATIONS:
- Indoor environment resembling a classroom or computer laboratory.
- Grey concrete floor.
- Multiple wooden tables with black metal frames.
- Computer desks with monitors; some screens appear blue.
- White background wall with vertically hanging banners or flags.
- Large green potted plant on the right side.
- Image is blurry, with apparent motion blur or camera movement.

LIMITATIONS:
- No identifiable individuals established.
- Exact location and activity cannot be determined from this frame.
- Image quality limits detailed examination.

EOF

┌──(kali㉿kali)-[~/SBT-DF204_CaseStudy5]
└─$
```

Figure D8.2. Source screenshot page 92: Recorded visual-examination notes for a sampled frame from DJI_0001.MP4.

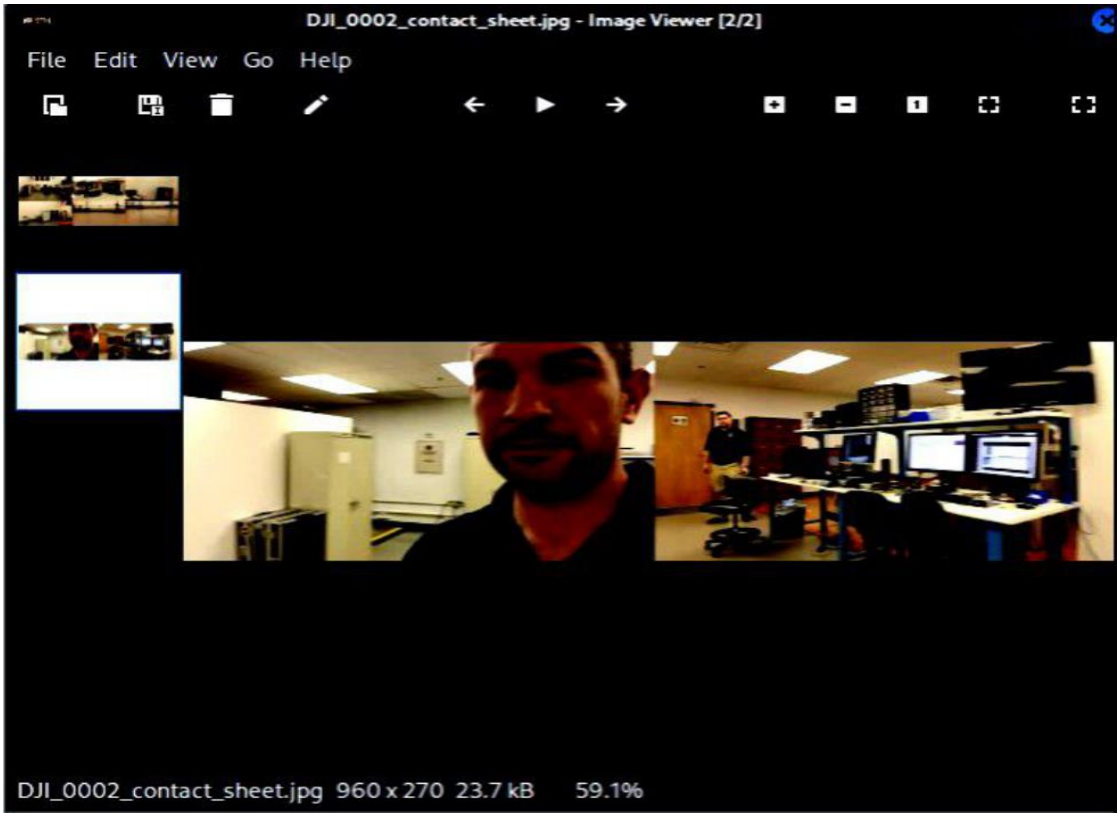


Figure D9.1. Source screenshot page 98: Sampled frame from DJI_0002.MP4 showing an indoor technical workspace andpeople in the scene.

```
┌──(kali㉿kali)-[~/SBT-DF204_CaseStudy5]
└─$ cat <<'EOF' >> logs/DF048_E01_visual_examination.txt

VIDEO: DJI_0002.MP4
SOURCE: EWF image DF048.E01
DURATION: Approximately 52.72 seconds
SAMPLED FRAMES: 2 (approximately 30-second interval)

VISUAL OBSERVATIONS:
- Frame 01 shows an indoor room with desks, computers and technical equipment.
- Frame 02 shows an adult man wearing a dark shirt in the foreground.
- A second person is visible standing near a doorway in the background.
- Multiple computer workstations and monitors are visible.
- Office chairs, storage cabinets and other equipment are present.
- The environment resembles a technical or computer-laboratory workspace.

FORENSIC SIGNIFICANCE:
- At least two people are visually present in the sampled footage.
- Their identities, roles and activities cannot be established.
- The exact building and geographical location remain undetermined.

LIMITATIONS:
- Only two frames sampled at approximately 30-second intervals were examined.
- This is not a complete examination of every video frame.

EOF
```

Figure D9.2. Source screenshot page 100: Visual-examination notes for DJI_0002.MP4, including the limits of the sampled-framereview.

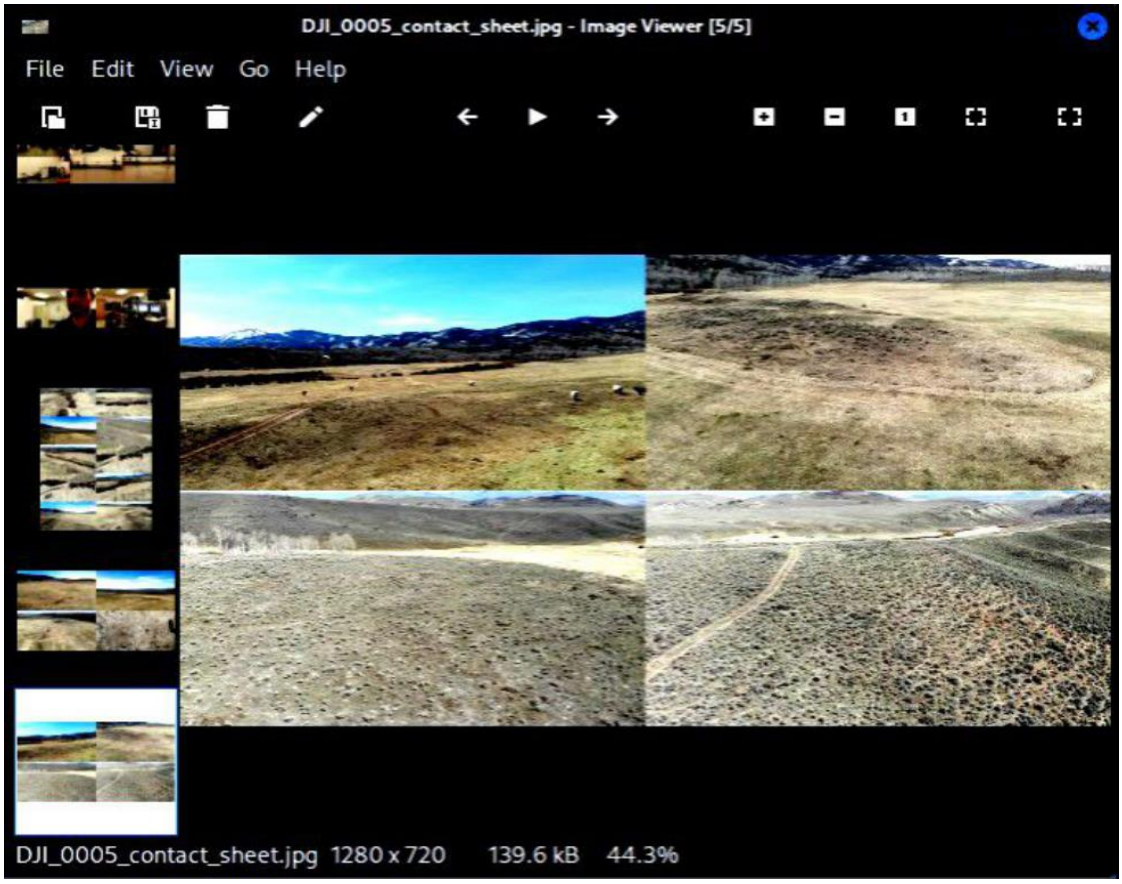


Figure D10.1. Source screenshot page 106: Sampled frames from DJI_0005.MP4 showing open, dry terrain and distant hills.

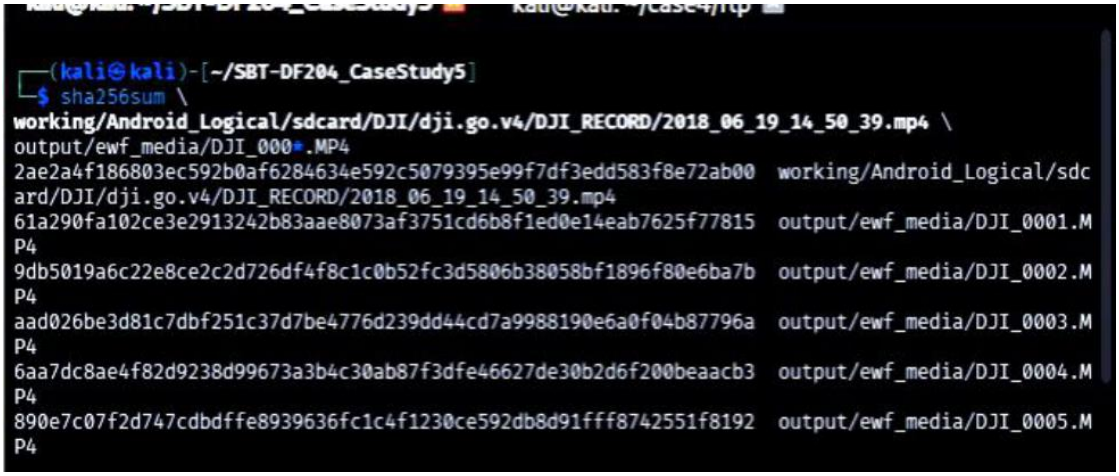


Figure D10.2. Source screenshot page 108: SHA-256 hash output comparing the Android cached video with the five extractedEWF video files.

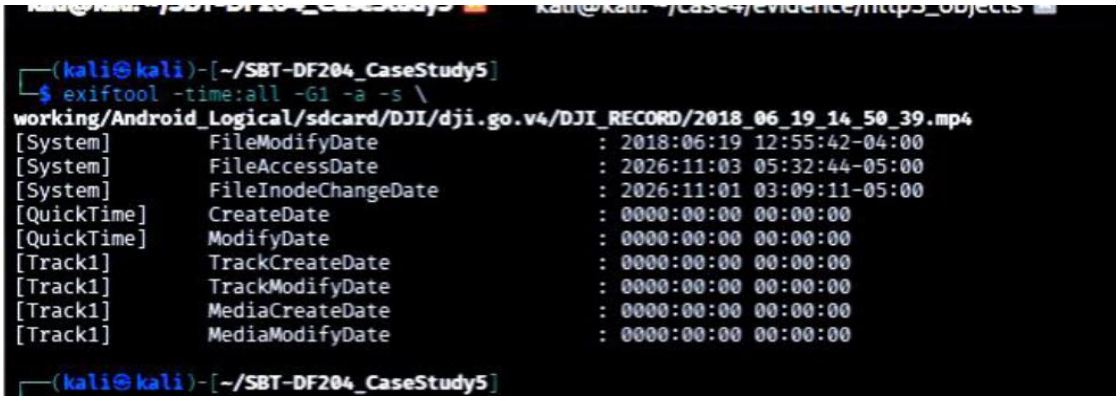


Figure D11.1. Source screenshot page 110: Metadata output for the Android cached video; filesystem and metadata timestampsrequire cautious interpretation.

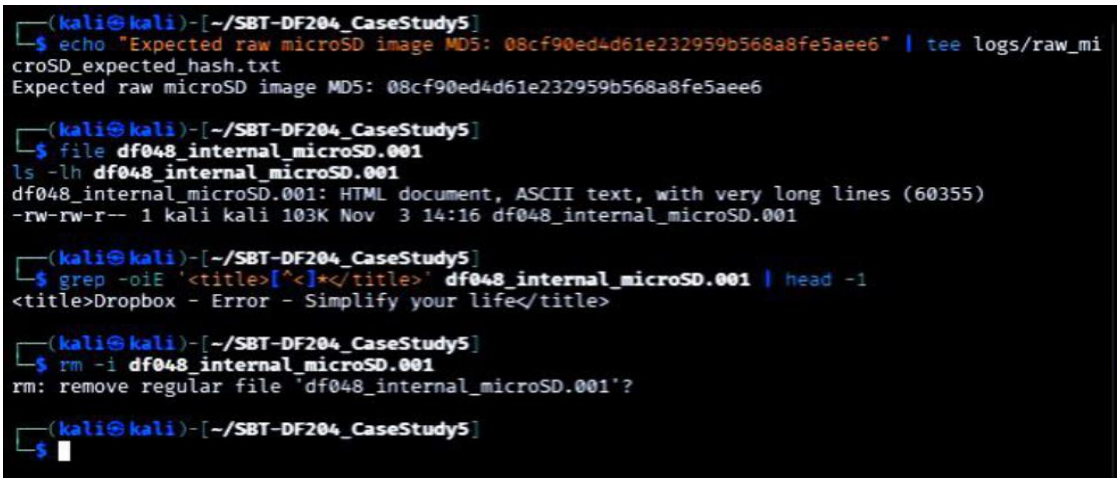


Figure D11.2. Source screenshot page 112: Failed raw microSD image download: the retrieved object is identified as an HTMLDropbox error page, not a forensic image.